Sri Lanka Institute of Information Technology

IT3021 - Data warehousing and

Business Intelligence

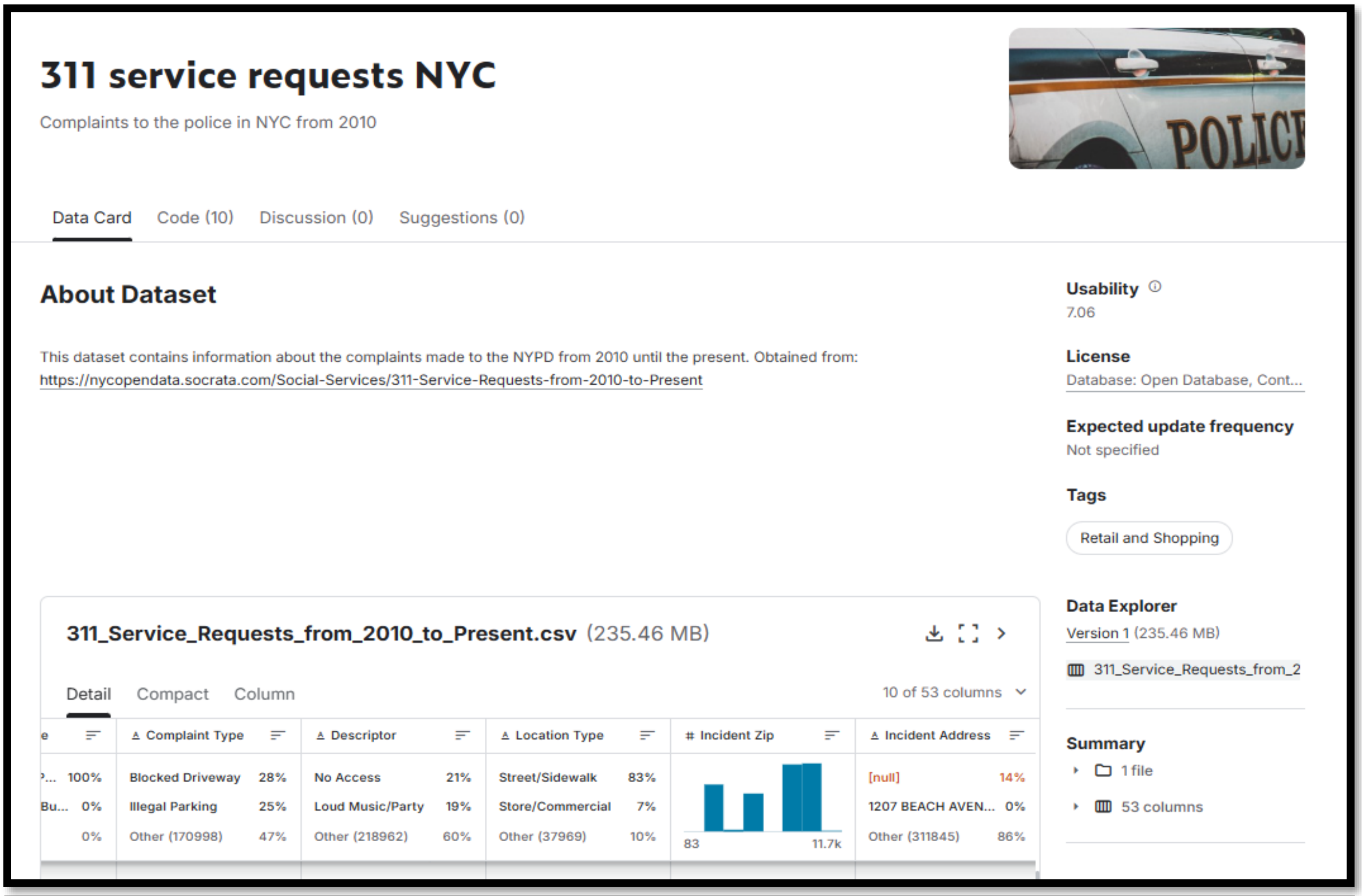
Assignment 01

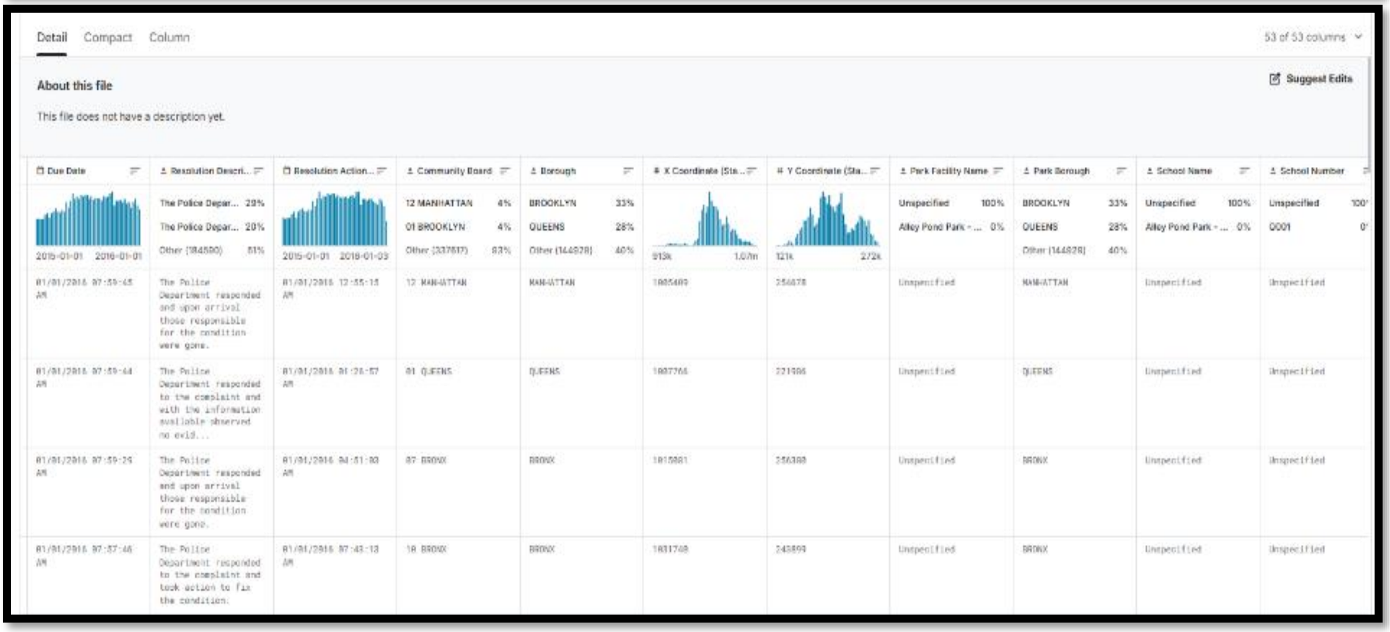
IT23672550

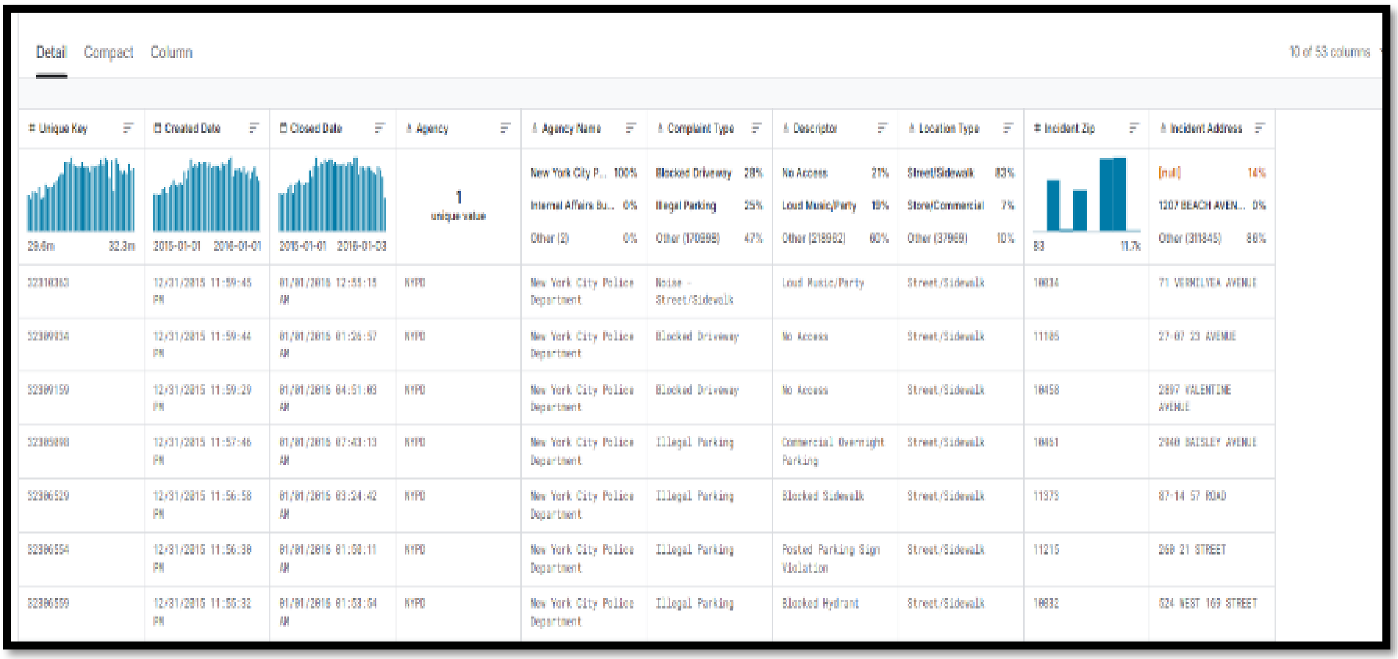
Wijesuriya W.M.S.N

<u>Task 1: Data set selection</u>

Dataset 👉 https://www.kaggle.com/datasets/pablomonleon/311-service-requests-nyc/data



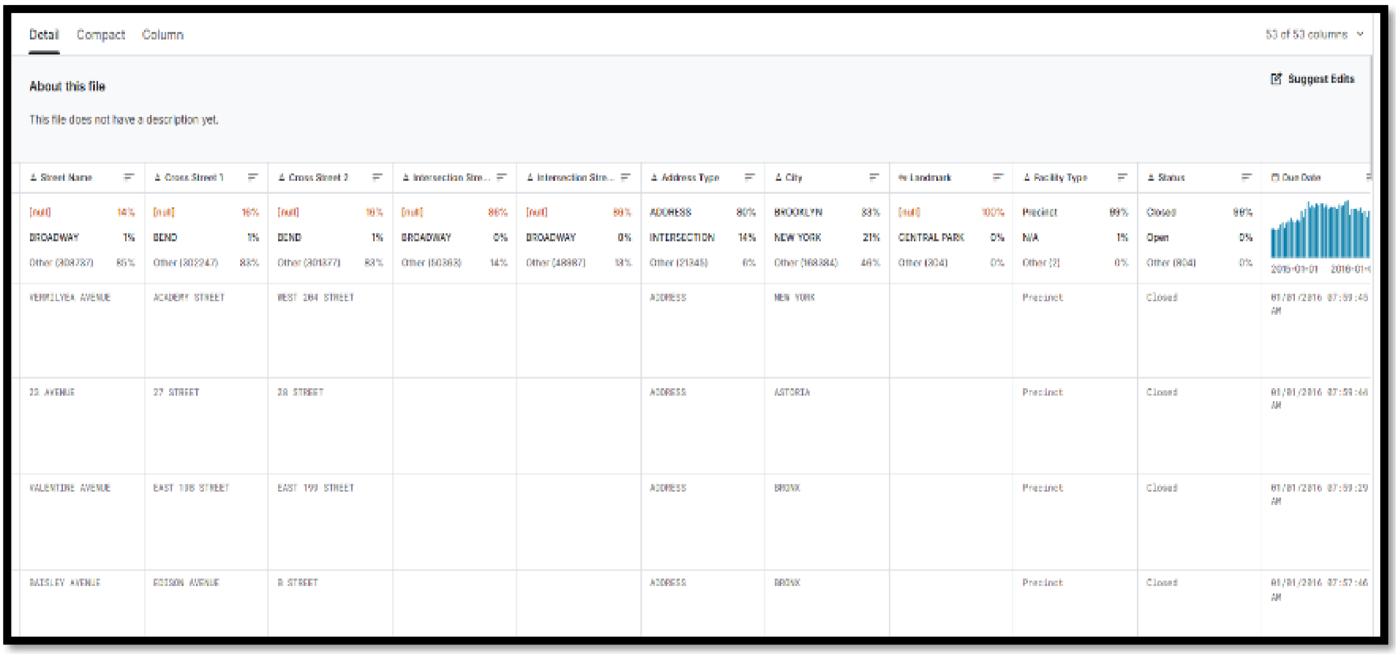




Detail Compact Column

10 of 53 columns

Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor	Location Type	Incident Zip	Incident Address
29.6m 32.3m	2015-01-01 2016-01-01	2015-01-01 2016-01-03	1 unique value	New York City P... 100% Internal Affairs Bu... 0% Other (2) 0%	Blocked Driveway 28% Illegal Parking 25% Other (170998) 47%	No Access 21% Loud Music/Party 19% Other (218962) 60%	Street/Sidewalk 83% Store/Commercial 7% Other (37969) 10%	83 11.7k	[null] 14% 1207 BEACH AVEN... 0% Other (311845) 86%
32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10034	71 VERMILYEA AVENUE
32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/Sidewalk	11105	27-07 23 AVENUE
32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/Sidewalk	10458	2897 VALENTINE AVENUE
32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking	Street/Sidewalk	10461	2940 BAISLEY AVENUE
32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk	Street/Sidewalk	11373	87-14 57 ROAD
32306554	12/31/2015 11:56:30 PM	01/01/2016 01:50:11 AM	NYPD	New York City Police Department	Illegal Parking	Posted Parking Sign Violation	Street/Sidewalk	11215	260 21 STREET
32306559	12/31/2015 11:55:32 PM	01/01/2016 01:53:54 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Hydrant	Street/Sidewalk	10032	524 WEST 169 STREET



Detail Compact Column

53 of 53 columns

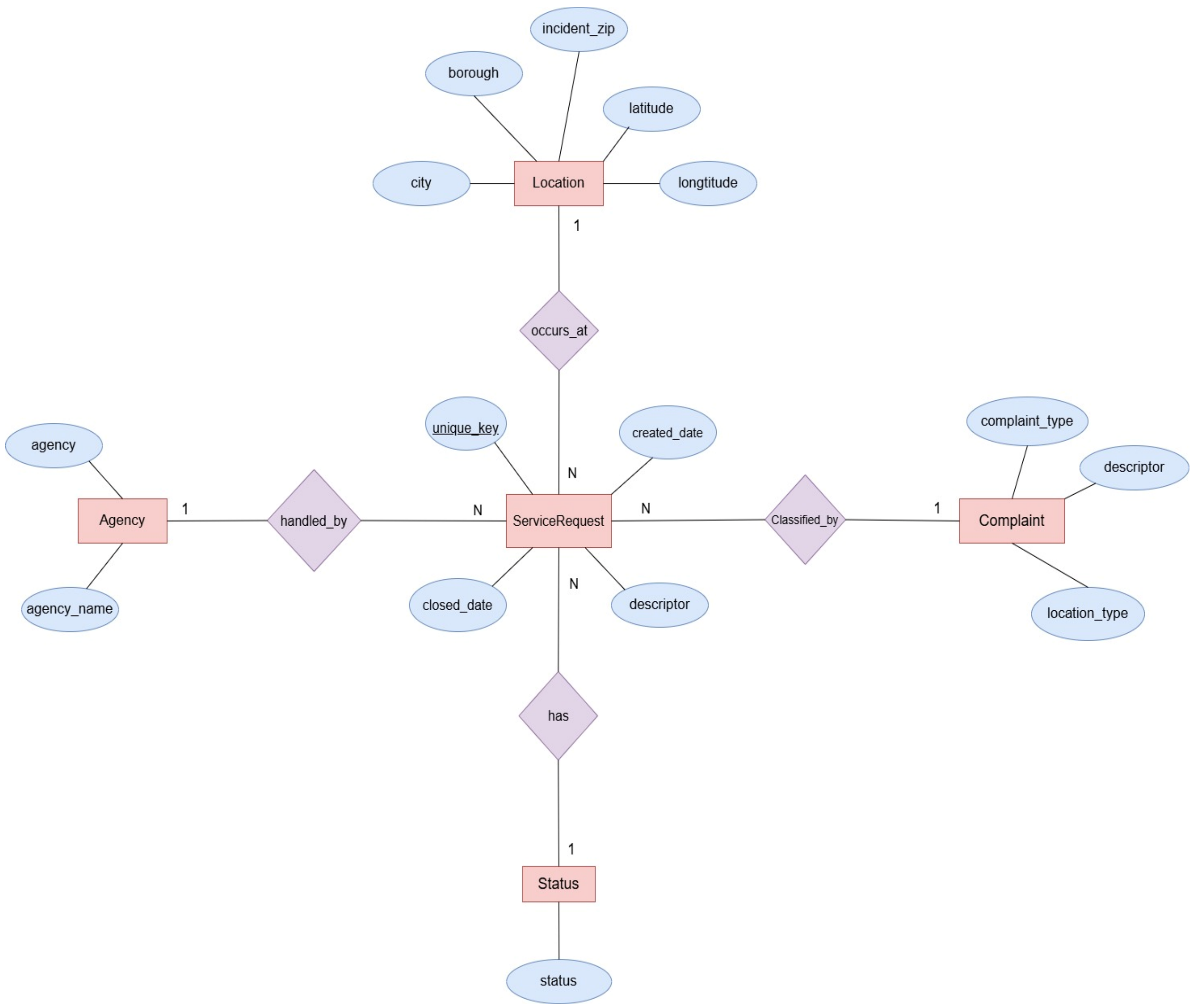
About this file

Suggest Edits

This file does not have a description yet.

Street Name	Cross Street 1	Cross Street 2	Intersection Stre...	Intersection Stre...	Address Type	City	Landmark	Facility Type	Status	Due Date
[null] 14% BROADWAY 1% Other (308787) 85%	[null] 16% BEND 1% Other (302247) 83%	[null] 16% BEND 1% Other (301377) 83%	[null] 86% BROADWAY 0% Other (50363) 14%	[null] 86% BROADWAY 0% Other (48987) 13%	ADDRESS 80% INTERSECTION 14% Other (21345) 6%	BROOKLYN 33% NEW YORK 21% Other (168384) 46%	[null] 100% CENTRAL PARK 0% Other (304) 0%	Precinct 99% N/A 1% Other (2) 0%	Closed 99% Open 0% Other (804) 0%	2015-01-01 2016-01-(
VERMILYEA AVENUE	ACADEMY STREET	WEST 204 STREET			ADDRESS	NEW YORK		Precinct	Closed	01/01/2016 07:59:45 AM
23 AVENUE	27 STREET	28 STREET			ADDRESS	ASTORIA		Precinct	Closed	01/01/2016 07:59:44 AM
VALENTINE AVENUE	EAST 198 STREET	EAST 199 STREET			ADDRESS	BRONX		Precinct	Closed	01/01/2016 07:59:29 AM
BAISLEY AVENUE	EDISON AVENUE	B STREET			ADDRESS	BRONX		Precinct	Closed	01/01/2016 07:57:46 AM

ER Diagram



Conceptual ER diagram for the NYC 311 service request dataset

1.1 Selected Dataset

The selected database for this assignment is the NYC 311 Service Requests dataset. This dataset contains records of public service requests submitted through the New York City 311 system. Each row represents a single service request and includes key operational details such as request creation date, closure date, responsible agency, complaint type, status, and location-related information.

The dataset is considered an OLTP-style dataset because it stores transactional records in a single wide table rather than being structured into fact and dimension tables. Therefore, it is suitable for transformation into a dimensional model for data warehousing and business intelligence purposes.

1.2 Justification for Dataset Selection

The selected dataset satisfies the assignment requirements and provides a meaningful real-world scenario. Unlike commonly used sales datasets, this dataset focuses on urban public service management, making it more unique and practical.

The dataset contains a large number of records collected over multiple years, ensuring sufficient data volume for analysis. It also includes more than 20 attributes, which allows the creation of multiple dimensions, hierarchies, and analytical measures.

Key attributes such as created date, closed date, complaint type, agency, and location fields (borough, city, zip code, latitude, and longitude) support dimensional modelling and enable the development of meaningful business intelligence reports.

Item	Description
Dataset Name	NYC 311 Service Requests
Source	Kaggle
Dataset Type	OLTP-style operational dataset
Main Entity	One service request
Main Time Fields	Created Date, Closed Date
Key Analysis Fields	Complaint Type, Agency, Borough, Status

1.3 Suitability for Data Warehousing and BI

The dataset is highly suitable for data warehouse development as it contains detailed operational data that can be transformed into a dimensional model.

The data can be organized into multiple dimensions such as Date, Agency, Complaint, Location, and Status, while the service request itself can be modelled as a central fact table. The presence of time-based attributes allows temporal analysis, while location and complaint-related fields support hierarchical and aggregated reporting.

Additionally, the dataset represents a clear lifecycle of events from request creation to closure, making it ideal for ETL processes, fact table loading, and performance analysis. This structure enables the implementation of OLAP cubes and advanced reporting features.

Task 2: Preparation of Data Sources

2.1 Source Types Used

To support ETL development and satisfy the assignment requirements, the dataset was prepared using multiple source types.

Three different source types were used in this project:

- **CSV File Source**: The original NYC 311 dataset was obtained as a CSV file containing raw service request data.
- **SQL Server Source**: The raw CSV data was imported into SQL Server into a table named raw_311_requests. A structured table named src_311_requests_sql was then created by selecting and cleaning relevant attributes.
- **Excel Lookup Source**: An Excel file named complaint_category_lookup.xlsx was created to enrich the dataset with additional complaint-related information such as complaint group, priority level, and service category.

This approach ensures the use of multiple data sources, as required by the assignment.

2.2 Information Available in Each Source

Each data source provides different types of information required for the ETL process:

- The **CSV file** contains the original raw dataset, including request ID, dates, complaint details, agency information, status, and location attributes.
- The **SQL Server source** contains clean and structured data that is more suitable for transformation and loading into the data warehouse.
- The **Excel lookup file** provides additional classification attributes such as complaint group, priority level, and service category, which are not available in the original dataset.

This separation of data sources improves data quality and supports dimensional modelling and hierarchy creation.

2.3 Justification for Source Preparation

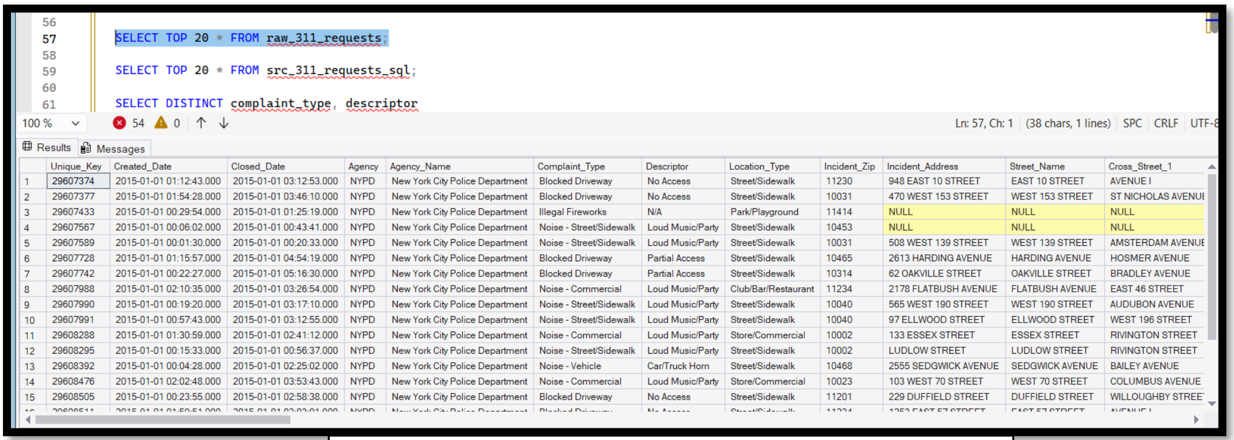
The following steps were performed to prepare the data sources:

1. The NYC 311 dataset was downloaded as a CSV file.
2. The dataset was examined to understand its structure and key attributes.
3. The CSV file was imported into SQL Server as a raw table named raw_311_requests.
4. A cleaned and structured table named src_311_requests_sql was created by selecting relevant columns and applying basic transformations.
5. An Excel lookup file (complaint_category_lookup.xlsx) was created to categorize complaints into groups, priority levels, and service categories.
6. Data validation checks were performed to ensure consistency and completeness across all sources.

The use of multiple data sources improves the flexibility and effectiveness of the ETL process. It allows the integration of raw operational data with enriched classification data, which enhances the analytical capabilities of the data warehouse.

This approach also enables the creation of hierarchical dimensions and supports more meaningful reporting. By separating the data into different formats such as CSV, SQL, and Excel, the system becomes more scalable and aligned with real-world data warehousing practices.

2.4 Evidence of Source Preparation



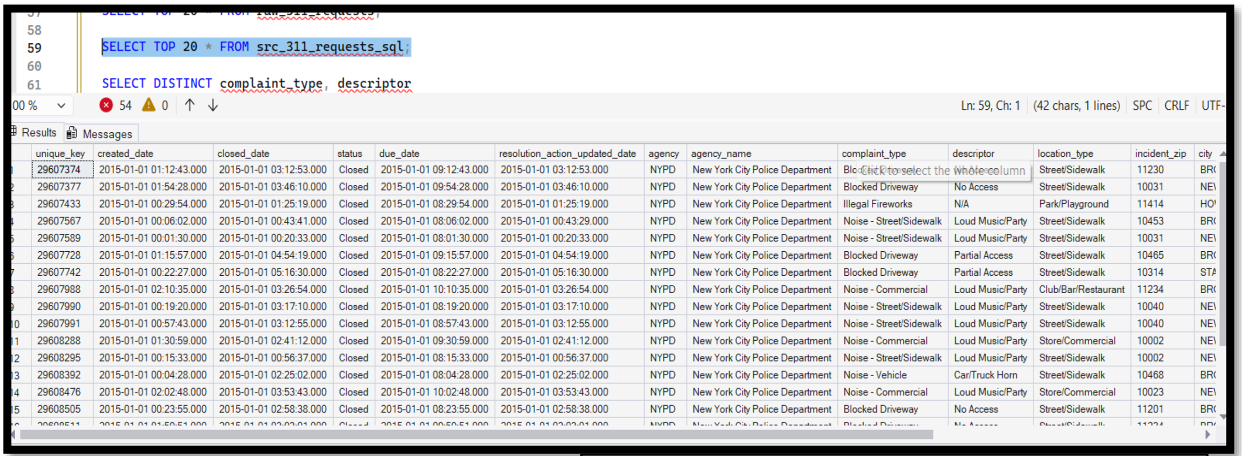
```
SELECT TOP 20 * FROM raw_311_requests;

SELECT TOP 20 * FROM src_311_requests_sql;

SELECT DISTINCT complaint_type, descriptor
```

100 % | 54 | 0 | Ln: 57, Ch: 1 | (38 chars, 1 lines) | SPC | CRLF | UTF-8

Results | Messages

	Unique_Key	Created_Date	Closed_Date	Agency	Agency_Name	Complaint_Type	Descriptor	Location_Type	Incident_Zip	Incident_Address	Street_Name	Cross_Street_1
1	29607374	2015-01-01 01:12:43.000	2015-01-01 03:12:53.000	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/Sidewalk	11230	948 EAST 10 STREET	EAST 10 STREET	AVENUE I
2	29607377	2015-01-01 01:54:28.000	2015-01-01 03:46:10.000	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/Sidewalk	10031	470 WEST 153 STREET	WEST 153 STREET	ST NICHOLAS AVENUE
3	29607433	2015-01-01 00:29:54.000	2015-01-01 01:25:19.000	NYPD	New York City Police Department	Illegal Fireworks	N/A	Park/Playground	11414	NULL	NULL	NULL
4	29607567	2015-01-01 00:06:02.000	2015-01-01 00:43:41.000	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10453	NULL	NULL	NULL
5	29607589	2015-01-01 00:01:30.000	2015-01-01 00:20:33.000	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10031	508 WEST 139 STREET	WEST 139 STREET	AMSTERDAM AVENUE
6	29607728	2015-01-01 01:15:57.000	2015-01-01 04:54:19.000	NYPD	New York City Police Department	Blocked Driveway	Partial Access	Street/Sidewalk	10465	2613 HARDING AVENUE	HARDING AVENUE	HOSMER AVENUE
7	29607742	2015-01-01 00:22:27.000	2015-01-01 05:16:30.000	NYPD	New York City Police Department	Blocked Driveway	Partial Access	Street/Sidewalk	10314	62 OAKVILLE STREET	OAKVILLE STREET	BRADLEY AVENUE
8	29607988	2015-01-01 02:10:35.000	2015-01-01 03:26:54.000	NYPD	New York City Police Department	Noise - Commercial	Loud Music/Party	Club/Bar/Restaurant	11234	2178 FLATBUSH AVENUE	FLATBUSH AVENUE	EAST 46 STREET
9	29607990	2015-01-01 00:19:20.000	2015-01-01 03:17:10.000	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10040	565 WEST 190 STREET	WEST 190 STREET	AUDUBON AVENUE
10	29607991	2015-01-01 00:57:43.000	2015-01-01 03:12:55.000	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10040	97 ELLWOOD STREET	ELLWOOD STREET	WEST 196 STREET
11	29608288	2015-01-01 01:30:59.000	2015-01-01 02:41:12.000	NYPD	New York City Police Department	Noise - Commercial	Loud Music/Party	Store/Commercial	10002	133 ESSEX STREET	ESSEX STREET	RIVINGTON STREET
12	29608295	2015-01-01 00:15:33.000	2015-01-01 00:56:37.000	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10002	LUDLOW STREET	LUDLOW STREET	RIVINGTON STREET
13	29608392	2015-01-01 00:04:28.000	2015-01-01 02:25:02.000	NYPD	New York City Police Department	Noise - Vehicle	Car/Truck Horn	Street/Sidewalk	10468	2555 SEDGWICK AVENUE	SEDGWICK AVENUE	BAILEY AVENUE
14	29608476	2015-01-01 02:02:48.000	2015-01-01 03:53:43.000	NYPD	New York City Police Department	Noise - Commercial	Loud Music/Party	Store/Commercial	10023	103 WEST 70 STREET	WEST 70 STREET	COLUMBUS AVENUE
15	29608505	2015-01-01 00:23:55.000	2015-01-01 02:58:38.000	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/Sidewalk	11201	229 DUFFIELD STREET	DUFFIELD STREET	WILLOUGHBY STREET

Raw NYC 311 data loaded into SQL Server (raw_311_requests)



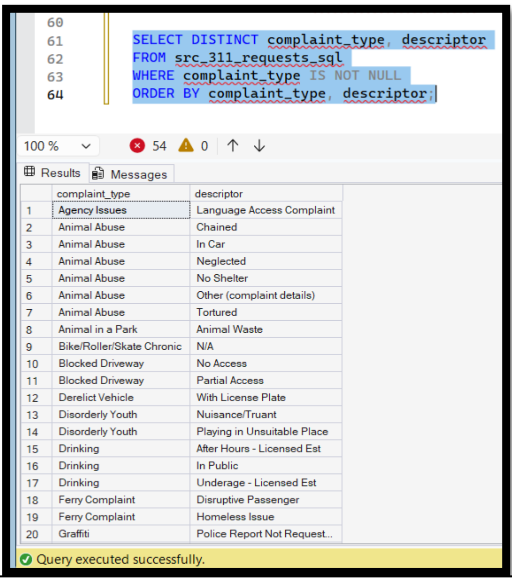
```
SELECT TOP 20 * FROM src_311_requests_sql;

SELECT DISTINCT complaint_type, descriptor
```

00 % | 54 | 0 | Ln: 59, Ch: 1 | (42 chars, 1 lines) | SPC | CRLF | UTF-

Results | Messages

	unique_key	created_date	closed_date	status	due_date	resolution_action_updated_date	agency	agency_name	complaint_type	descriptor	location_type	incident_zip	city
1	29607374	2015-01-01 01:12:43.000	2015-01-01 03:12:53.000	Closed	2015-01-01 09:12:43.000	2015-01-01 03:12:53.000	NYPD	New York City Police Department	Blo[illegible]	[illegible]	Street/Sidewalk	11230	BR
2	29607377	2015-01-01 01:54:28.000	2015-01-01 03:46:10.000	Closed	2015-01-01 09:54:28.000	2015-01-01 03:46:10.000	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/Sidewalk	10031	NE
3	29607433	2015-01-01 00:29:54.000	2015-01-01 01:25:19.000	Closed	2015-01-01 08:29:54.000	2015-01-01 01:25:19.000	NYPD	New York City Police Department	Illegal Fireworks	N/A	Park/Playground	11414	HO
4	29607567	2015-01-01 00:06:02.000	2015-01-01 00:43:41.000	Closed	2015-01-01 08:06:02.000	2015-01-01 00:43:29.000	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10453	BR
5	29607589	2015-01-01 00:01:30.000	2015-01-01 00:20:33.000	Closed	2015-01-01 08:01:30.000	2015-01-01 00:20:33.000	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10031	NE
6	29607728	2015-01-01 01:15:57.000	2015-01-01 04:54:19.000	Closed	2015-01-01 09:15:57.000	2015-01-01 04:54:19.000	NYPD	New York City Police Department	Blocked Driveway	Partial Access	Street/Sidewalk	10465	BR
7	29607742	2015-01-01 00:22:27.000	2015-01-01 05:16:30.000	Closed	2015-01-01 08:22:27.000	2015-01-01 05:16:30.000	NYPD	New York City Police Department	Blocked Driveway	Partial Access	Street/Sidewalk	10314	STA
8	29607988	2015-01-01 02:10:35.000	2015-01-01 03:26:54.000	Closed	2015-01-01 10:10:35.000	2015-01-01 03:26:54.000	NYPD	New York City Police Department	Noise - Commercial	Loud Music/Party	Club/Bar/Restaurant	11234	BR
9	29607990	2015-01-01 00:19:20.000	2015-01-01 03:17:10.000	Closed	2015-01-01 08:19:20.000	2015-01-01 03:17:10.000	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10040	NE
10	29607991	2015-01-01 00:57:43.000	2015-01-01 03:12:55.000	Closed	2015-01-01 08:57:43.000	2015-01-01 03:12:55.000	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10040	NE
11	29608288	2015-01-01 01:30:59.000	2015-01-01 02:41:12.000	Closed	2015-01-01 09:30:59.000	2015-01-01 02:41:12.000	NYPD	New York City Police Department	Noise - Commercial	Loud Music/Party	Store/Commercial	10002	NE
12	29608295	2015-01-01 00:15:33.000	2015-01-01 00:56:37.000	Closed	2015-01-01 08:15:33.000	2015-01-01 00:56:37.000	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/Sidewalk	10002	NE
13	29608392	2015-01-01 00:04:28.000	2015-01-01 02:25:02.000	Closed	2015-01-01 08:04:28.000	2015-01-01 02:25:02.000	NYPD	New York City Police Department	Noise - Vehicle	Car/Truck Horn	Street/Sidewalk	10468	BR
14	29608476	2015-01-01 02:02:48.000	2015-01-01 03:53:43.000	Closed	2015-01-01 10:02:48.000	2015-01-01 03:53:43.000	NYPD	New York City Police Department	Noise - Commercial	Loud Music/Party	Store/Commercial	10023	NE
15	29608505	2015-01-01 00:23:55.000	2015-01-01 02:58:38.000	Closed	2015-01-01 08:23:55.000	2015-01-01 02:58:38.000	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/Sidewalk	11201	BR

Cleaned and structured dataset (src_311_requests_sql)



```
SELECT DISTINCT complaint_type, descriptor
FROM src_311_requests_sql
WHERE complaint_type IS NOT NULL
ORDER BY complaint_type, descriptor;
```

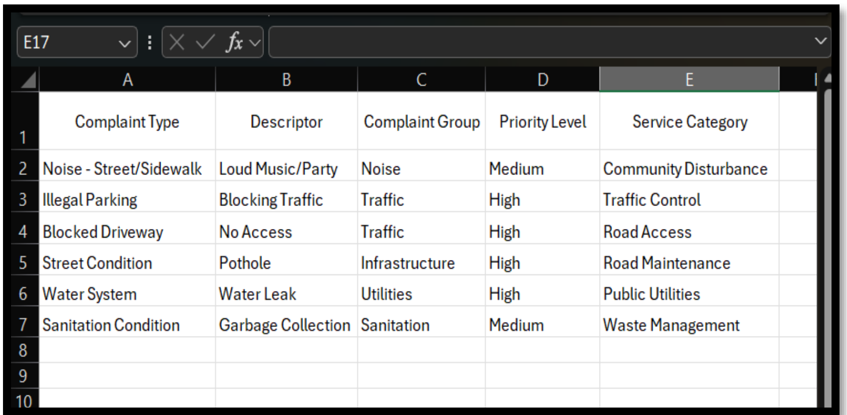
100 % | 54 | 0

Results | Messages

	complaint_type	descriptor
1	Agency Issues	Language Access Complaint
2	Animal Abuse	Chained
3	Animal Abuse	In Car
4	Animal Abuse	Neglected
5	Animal Abuse	No Shelter
6	Animal Abuse	Other (complaint details)
7	Animal Abuse	Tortured
8	Animal in a Park	Animal Waste
9	Bike/Roller/Skate Chronic	N/A
10	Blocked Driveway	No Access
11	Blocked Driveway	Partial Access
12	Derelict Vehicle	With License Plate
13	Disorderly Youth	Nuisance/Truant
14	Disorderly Youth	Playing in Unsuitable Place
15	Drinking	After Hours - Licensed Est
16	Drinking	In Public
17	Drinking	Underage - Licensed Est
18	Ferry Complaint	Disruptive Passenger
19	Ferry Complaint	Homeless Issue
20	Graffiti	Police Report Not Request...

Query executed successfully.

Distinct complaint types extracted from source data



E17

	A	B	C	D	E
1	Complaint Type	Descriptor	Complaint Group	Priority Level	Service Category
2	Noise - Street/Sidewalk	Loud Music/Party	Noise	Medium	Community Disturbance
3	Illegal Parking	Blocking Traffic	Traffic	High	Traffic Control
4	Blocked Driveway	No Access	Traffic	High	Road Access
5	Street Condition	Pothole	Infrastructure	High	Road Maintenance
6	Water System	Water Leak	Utilities	High	Public Utilities
7	Sanitation Condition	Garbage Collection	Sanitation	Medium	Waste Management
8					
9					
10					

Excel lookup file used for complaint categorization

CSV file

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	Unique Key	Created Da	Closed Dat	Agency	Agency Na	Complaint	Descriptor	Location Ty	Incident Zi	Incident Ad	Street Nam	Cross Stre	Cross Stre	Intersectio	Intersectio	Address Ty	City	Landmark	Facility Typ	Status	Due Date	Resolution	Resolution
2	32310363	########	########	NYPD	New York (	Noise - Str	Loud Musi	Street/Side	10034	71 VERMIL	VERMILYEA	ACADEMY	WEST 204 STREET			ADDRESS	NEW YORK		Precinct	Closed	########	The Police	########
3	32309934	########	########	NYPD	New York (	Blocked Dr	No Access	Street/Side	11105	27-07 23 A	23 AVENU	27 STREET	28 STREET			ADDRESS	ASTORIA		Precinct	Closed	########	The Police	########
4	32309159	########	########	NYPD	New York (	Blocked Dr	No Access	Street/Side	10458	2897 VALE	VALENTIN	EAST 198 S	EAST 199 STREET			ADDRESS	BRONX		Precinct	Closed	########	The Police	########
5	32305098	########	########	NYPD	New York (	Illegal Park	Commerci	Street/Side	10461	2940 BAISL	BAISLEY AV	EDISON AV	B STREET			ADDRESS	BRONX		Precinct	Closed	########	The Police	########
6	32306529	########	########	NYPD	New York (	Illegal Park	Blocked Si	Street/Side	11373	87-14 57 R	57 ROAD	SEABURY S	HOFFMAN DRIVE			ADDRESS	ELMHURST		Precinct	Closed	########	The Police	########
7	32306554	########	########	NYPD	New York (	Illegal Park	Posted Par	Street/Side	11215	260 21 STR	21 STREET	5 AVENUE	6 AVENUE			ADDRESS	BROOKLYN		Precinct	Closed	########	The Police	########
8	32306559	########	########	NYPD	New York (	Illegal Park	Blocked Hy	Street/Side	10032	524 WEST	WEST 169 S	AMSTERDA	AUDUBON AVENUE			ADDRESS	NEW YORK		Precinct	Closed	########	The Police	########
9	32307009	########	########	NYPD	New York (	Blocked Dr	No Access	Street/Side	10457	501 EAST 1	EAST 171 S	WASHINGT	3 AVENUE			ADDRESS	BRONX		Precinct	Closed	########	The Police	########
10	32308581	########	########	NYPD	New York (	Illegal Park	Posted Par	Street/Side	11415	83-44 LEFF	LEFFERTS	BEND	BEND			ADDRESS	KEW GARDENS		Precinct	Closed	########	The Police	########
11	32308391	########	########	NYPD	New York (	Blocked Dr	No Access	Street/Side	11219	1408 66 ST	66 STREET	14 AVENU	NEW UTRECHT AVENUE			ADDRESS	BROOKLYN		Precinct	Closed	########	The Police	########
12	32305071	########	########	NYPD	New York (	Blocked Dr	No Access	Street/Side	11372	34-06 73 S	73 STREET	34 AVENU	35 AVENUE			ADDRESS	JACKSON HEIGHTS		Precinct	Closed	########	The Police	########
13	32306260	########	########	NYPD	New York (	Blocked Dr	No Access	Street/Side	10453	1770 UNDI	UNDERCLI	WEST 176 S	SEDGWICK AVENUE			ADDRESS	BRONX		Precinct	Closed	########	The Police	########
14	32306612	########	########	NYPD	New York (	Noise - Str	Loud Musi	Street/Side	10461	1701 PILGI	PILGRIM A	ROBERTS A	WESTCHESTER AVENUE			ADDRESS	BRONX		Precinct	Closed	########	The Police	########
15	32305074	########	########	NYPD	New York (	Illegal Park	Posted Par	Street/Side	11208	38 COX PL	COX PLAC	CRESCENT	DEAD END			ADDRESS	BROOKLYN		Precinct	Closed	########	The Police	########
16	32309424	########	########	NYPD	New York (	Derelict Ve	With Licen	Street/Side	11379	62-13 62 A	62 AVENU	62 STREET	64 STREET			ADDRESS	MIDDLE VILLAGE		Precinct	Closed	########	The Police	########
17	32309853	########	########	NYPD	New York (	Blocked Dr	Partial Acc	Street/Side	11374	61-34 AUS	AUSTIN ST	ELIOT AVE	62 AVENUE			ADDRESS	REGO PARK		Precinct	Closed	########	The Police	########
18	32305538	########	########	NYPD	New York (	Blocked Dr	No Access	Street/Side	11412	192-05 LIN	LINDEN BO	192 STREE	193 STREET			ADDRESS	SAINT ALBANS		Precinct	Closed	########	The Police	########
19	32310273	########	########	NYPD	New York (	Noise - Co	Loud Musi	Club/Bar/	11217	622 DEGRA	DEGRAW S	3 AVENUE	4 AVENUE			ADDRESS	BROOKLYN		Precinct	Closed	########	The Police	########
20	32306617	########	########	NYPD	New York (	Noise - Co	Loud Musi	Club/Bar/	11234	2192 FLAT	FLATBUSH	EAST 46 ST	AVENUE O			ADDRESS	BROOKLYN		Precinct	Closed	########	The Police	########
21	32308195	########	########	NYPD	New York (	Noise - Str	Loud Musi	Street/Side	10026	264 WEST	WEST 118 S	ST NICHOL	8 AVENUE			ADDRESS	NEW YORK		Precinct	Closed	########	The Police	########
22	32310127	########	########	NYPD	New York (	Illegal Park	Unauthoriz	Street/Side	10456	1417 WEBS	WEBSTER A	EAST 170 S	EAST 171 STREET			ADDRESS	BRONX		Precinct	Closed	########	The Police	########
23	32307994	########	########	NYPD	New York (	Illegal Park	Posted Par	Street/Side	11379	78-42 MET	METROPO	79 STREET	79 STREET			ADDRESS	MIDDLE VILLAGE		Precinct	Closed	########	The Police	########
24	32307233	########	########	NYPD	New York (	Noise - Co	Loud Musi	Store/Com	11234	3622 QUEN	QUENTIN R	EAST 36 ST	EAST 37 STREET			ADDRESS	BROOKLYN		Precinct	Closed	########	The Police	########
25	32308765	########	########	NYPD	New York (	Illegal Park	Double Par	Street/Side	10030	133 WEST	WEST 134 S	LENOX AVE	7 AVENUE			ADDRESS	NEW YORK		Precinct	Closed	########	The Police	########
26	32308423	########	########	NYPD	New York (	Blocked Dr	No Access	Street/Side	10467	3025 WALL	WALLACE A	ADEE AVEN	BURKE AVENUE			ADDRESS	BRONX		Precinct	Closed	########	The Police	########

311_Service_Requests_from_2010_ +

		W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	A
1	n	Resolution	Communit	Borough	X Coordina	Y Coordina	Park Facili	Park Borou	School Na	School Nu	School Reg	School Coc	School Phc	School Adc	School City	School Sta	School Zip	School Not	School or (	Vehicle Ty	Taxi Comp	Taxi Pick U	Bridge Hig	Bridge Hig	Road
2	e	########	12 MANHA	MANHATTA	1005409	254678	Unspecifie	MANHATTA	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
3	e	########	01 QUEEN	QUEENS	1007766	221986	Unspecifie	QUEENS	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
4	e	########	07 BRONX	BRONX	1015081	256380	Unspecifie	BRONX	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
5	e	########	10 BRONX	BRONX	1031740	243899	Unspecifie	BRONX	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
6	e	########	04 QUEEN	QUEENS	1019123	206375	Unspecifie	QUEENS	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
7	e	########	07 BROOK	BROOKLYN	986312	180032	Unspecifie	BROOKLYN	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
8	e	########	12 MANHA	MANHATTA	1001578	245627	Unspecifie	MANHATTA	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
9	e	########	03 BRONX	BRONX	1011117	244417	Unspecifie	BRONX	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
10	e	########	09 QUEEN	QUEENS	1030662	196163	Unspecifie	QUEENS	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
11	e	########	11 BROOK	BROOKLYN	984378	166541	Unspecifie	BROOKLYN	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
12	e	########	03 QUEEN	QUEENS	1013795	213487	Unspecifie	QUEENS	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
13	e	########	05 BRONX	BRONX	1006383	249532	Unspecifie	BRONX	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
14	e	########	10 BRONX	BRONX	1030293	247376	Unspecifie	BRONX	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
15	e	########	05 BROOK	BROOKLYN	1019054	189780	Unspecifie	BROOKLYN	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
16	e	########	05 QUEEN	QUEENS	1012143	199456	Unspecifie	QUEENS	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
17	e	########	06 QUEEN	QUEENS	1020165	205145	Unspecifie	QUEENS	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
18	e	########	12 QUEEN	QUEENS	1051009	191672	Unspecifie	QUEENS	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
19	e	########	06 BROOK	BROOKLYN	988846	186711	Unspecifie	BROOKLYN	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
20	e	########	18 BROOK	BROOKLYN	1003628	163910	Unspecifie	BROOKLYN	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
21	e	########	10 MANHA	MANHATTA	997164	232661	Unspecifie	MANHATTA	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
22	e	########	04 BRONX	BRONX	1009955	244133	Unspecifie	BRONX	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
23	e	########	05 QUEEN	QUEENS	1019142	199139	Unspecifie	QUEENS	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
24	e	########	18 BROOK	BROOKLYN	1001861	163265	Unspecifie	BROOKLYN	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
25	e	########	10 MANHA	MANHATTA	1000121	235845	Unspecifie	MANHATTA	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							
26	e	########	12 BRONX	BRONX	1021775	256294	Unspecifie	BRONX	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	Unspecifie	N							

311_Service_Requests_from_2010_ +

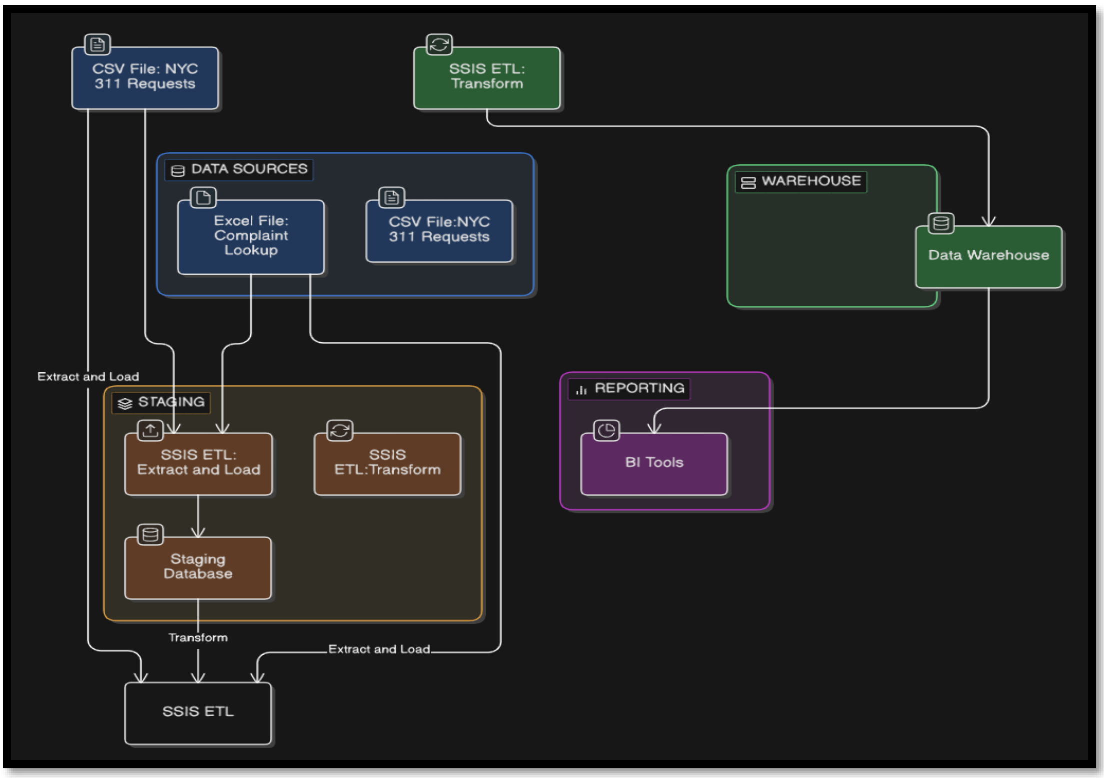
Task 3: Solution architecture

3.1 Overview of the Architecture

A data warehouse and business intelligence solution architecture was designed to integrate multiple data sources and support analytical reporting.

The architecture consists of several key layers, including data sources, staging area, ETL processes, data warehouse, and reporting layer. This layered approach ensures efficient data extraction, transformation, and loading while maintaining data quality and consistency.

3.2 Architecture Diagram



3.3 Components of the Architecture

- **Data Sources**
  The system uses multiple data sources, including a CSV file containing raw service request data, an SQL Server database containing structured data, and an Excel lookup file used for data enrichment. These sources provide both operational and supplementary information required for analysis.

- **Staging Area**
  The staging area is an intermediate storage layer where raw data from different sources is temporarily stored before transformation. This layer helps in isolating source data from the data warehouse and allows data cleaning and validation processes to be performed efficiently.

- **ETL Layer (SSIS)**
  The ETL (Extract, Transform, Load) layer is implemented using SQL Server Integration Services (SSIS). In this layer, data is extracted from multiple sources, transformed into a consistent format, enriched using lookup tables, and loaded into the data warehouse.
  Key transformations include data type conversions, filtering, lookup operations, and the creation of derived attributes.

- **Data Warehouse**
  The data warehouse stores the processed data in a structured dimensional model. A star schema is used, consisting of a central fact table and multiple dimension tables. This structure supports efficient querying and analysis.

- **Reporting and Analytics Layer**
  The final layer enables business intelligence and reporting. The data warehouse can be connected to tools such as SQL Server Analysis Services (SSAS), Power BI, or other reporting tools to generate insights, dashboards, and analytical reports.

3.4 Data Flow Explanation

The overall data flow begins with extracting data from multiple sources into the staging area. The ETL process then cleans and transforms the data before loading it into the data warehouse. Finally, the processed data is used for reporting and analysis.

Task 4: Data warehouse design & development

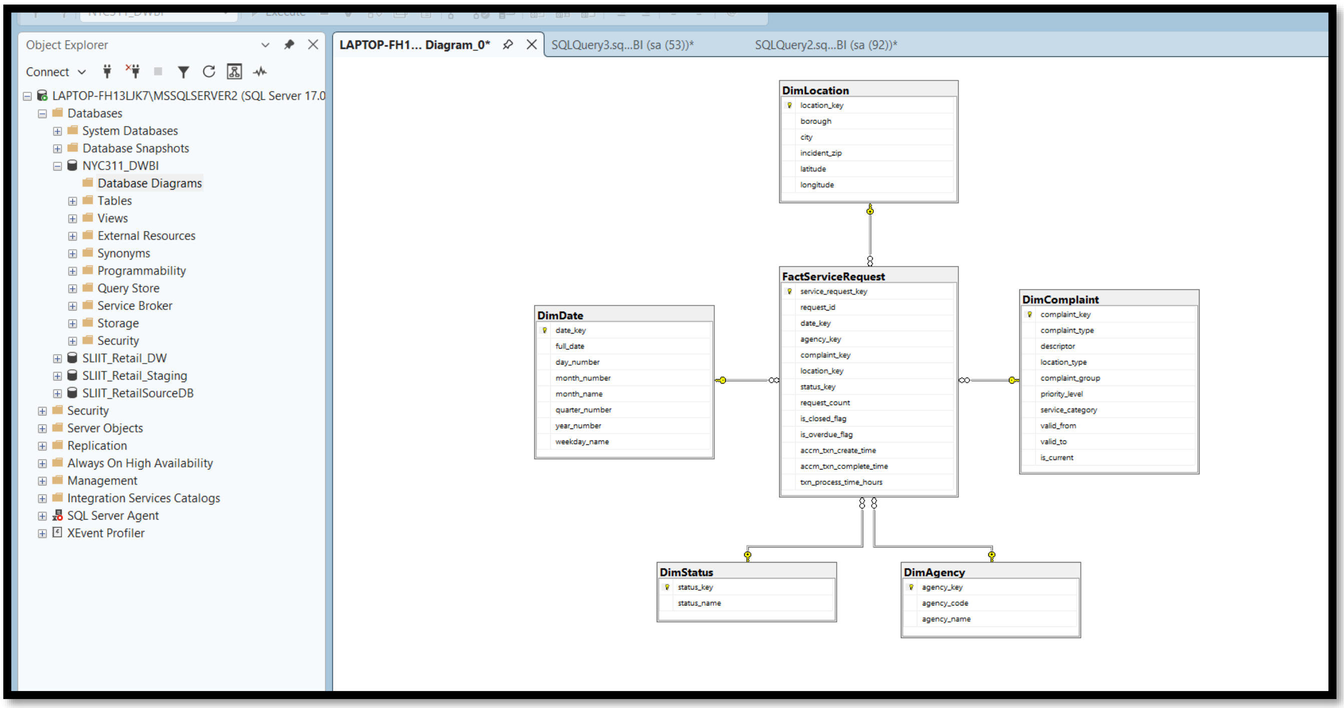
4.1 Dimensional Model Selection

A star schema was selected for the data warehouse design due to its simplicity and efficiency in analytical processing. In this model, a central fact table is connected to multiple dimension tables, enabling fast query performance and easy data interpretation.

This approach is suitable for the NYC 311 dataset as it supports analysis across different business perspectives such as time, agency, complaint type, location, and service request status.

4.2 Fact Table

The main fact table in the warehouse is FactServiceRequest. The grain of this fact table is defined as one row per NYC 311 service request. This means that each row represents one individual service request event from the source data. The fact table contains the natural request identifier together with foreign keys that link to the related dimensions. It also includes analytical fields such as request count, closed flag, overdue flag, and processing time attributes.



Measures Description

The fact table contains the following analytical measures:

- **request_count**: Represents each service request (used for aggregation)
- **is_closed_flag**: Indicates whether the request has been completed
- **is_overdue_flag**: Indicates whether the request exceeded expected processing time
- **txn_process_time_hours**: Represents the time taken to complete the request

Accumulating Fact Columns

To support lifecycle tracking, the fact table includes:

- **accm_txn_create_time**: Timestamp when the record is created
- **accm_txn_complete_time**: Timestamp when the request is completed

These attributes enable tracking of the request processing lifecycle and support accumulating fact table implementation.

Foreign Key Relationships

The fact table is linked to dimension tables using foreign keys:

- date_key -> DimDate
- agency_key -> DimAgency
- complaint_key -> DimComplaint
- location_key -> DimLocation
- status_key -> DimStatus

Additionally, the attribute unique_key represents the natural identifier of the service request.

4.3 Dimension Tables

Five-dimension tables were created in the warehouse: DimDate, DimAgency, DimComplaint, DimLocation, and DimStatus. These dimensions were chosen based on the most important descriptive attributes found in the source dataset. DimDate supports time-based analysis, DimAgency supports analysis by responsible agency, DimComplaint stores complaint-related descriptive information, DimLocation captures geographic details, and DimStatus stores the request status.

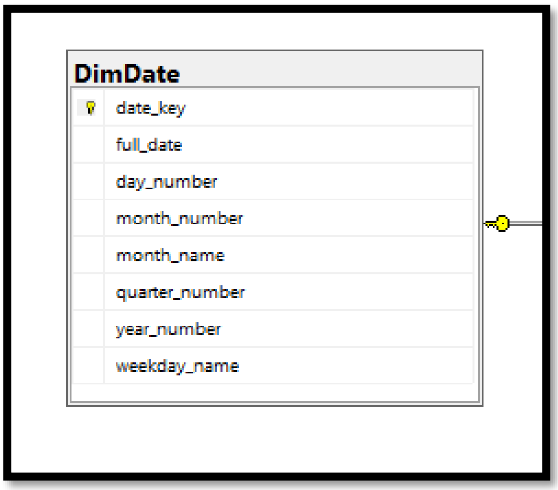
DimDate

This dimension supports time-based analysis.

Attributes include:

- date_key
- full_date
- day_number
- month_number
- month_name
- quarter_number
- year_number
- weekday_name

Column Name	Data Type	Allow Nulls
date_key	int	☐
full_date	date	☐
day_number	int	☑
month_number	int	☑
month_name	nvarchar(20)	☑
quarter_number	int	☑
year_number	int	☑
weekday_name	nvarchar(20)	☑
		☐

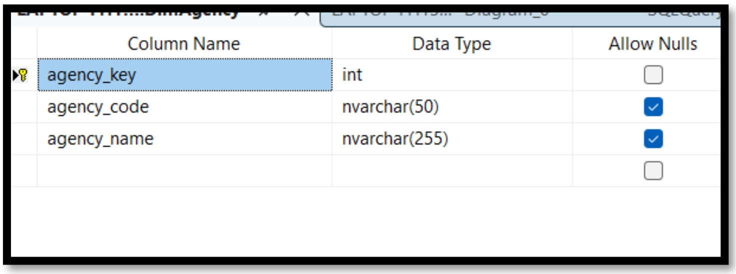


DimAgency

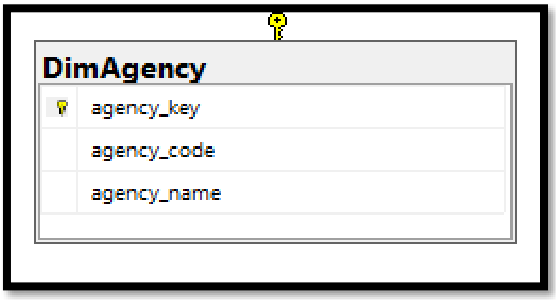
This dimension stores information about agencies handling service requests.

Attributes include:

- agency_key
- agency_code
- agency_name



Column Name	Data Type	Allow Nulls
agency_key	int	☐
agency_code	nvarchar(50)	☑
agency_name	nvarchar(255)	☑
		☐

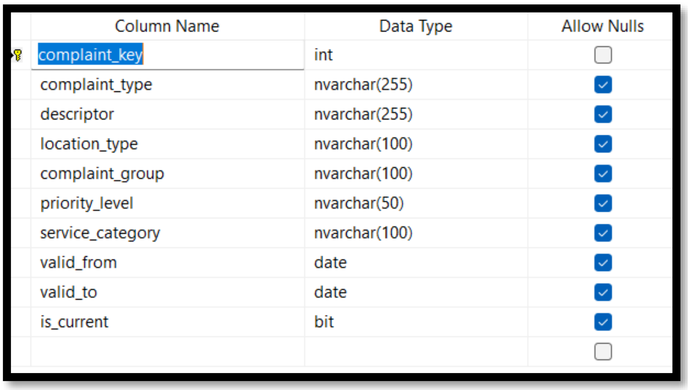


DimComplaint

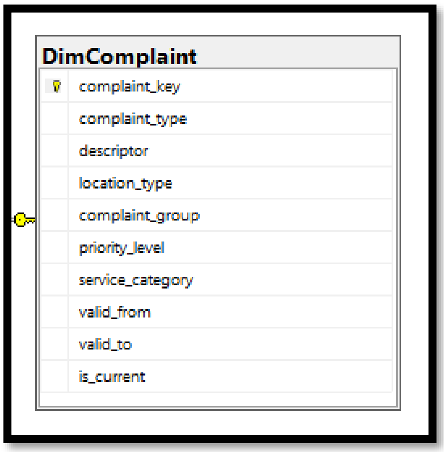
This dimension stores complaint-related attributes enriched using the Excel lookup source.

Attributes include:

- complaint_key
- complaint_type
- descriptor
- location_type
- complaint_group
- priority_level
- service_category



Column Name	Data Type	Allow Nulls
complaint_key	int	☐
complaint_type	nvarchar(255)	☑
descriptor	nvarchar(255)	☑
location_type	nvarchar(100)	☑
complaint_group	nvarchar(100)	☑
priority_level	nvarchar(50)	☑
service_category	nvarchar(100)	☑
valid_from	date	☑
valid_to	date	☑
is_current	bit	☑
		☐



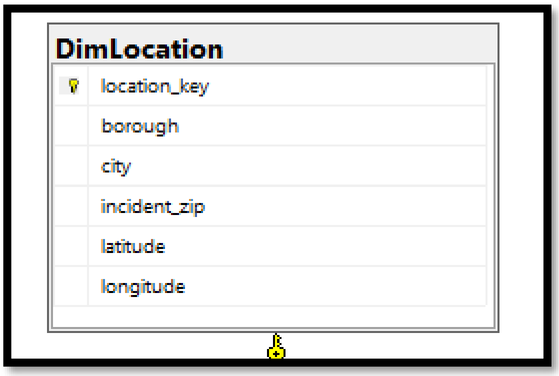
DimLocation

This dimension stores geographical information.

Attributes include:

- location_key
- borough
- city
- incident_zip
- latitude
- longitude

Column Name	Data Type	Allow Nulls
location_key	int	☐
borough	nvarchar(100)	☑
city	nvarchar(100)	☑
incident_zip	nvarchar(20)	☑
latitude	float	☑
longitude	float	☑



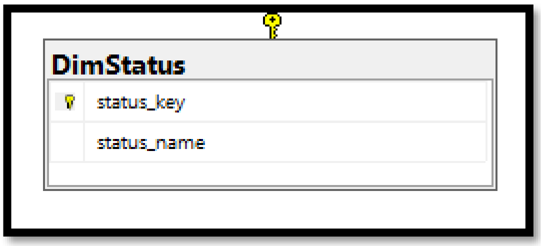
DimStatus

This dimension stores request status information.

Attributes include:

- status_key
- status_name

Column Name	Data Type	Allow Nulls
status_key	int	☐
status_name	nvarchar(100)	☑
		☐



4.4 Slowly Changing Dimension

The DimComplaint table is implemented as a Slowly Changing Dimension Type 2 (SCD Type 2).

This is required because complaint classifications such as complaint group, priority level, and service category may change over time. Instead of overwriting existing values, new records are inserted to preserve historical data.

The following attributes are used:

- **valid_from**: Start date of the record
- **valid_to**: End date of the record
- **is_current**: Indicates whether the record is active

This ensures accurate historical tracking and supports time-based analysis.

4.5 Hierarchies

Hierarchies were defined to support drill-down and roll-up operations in reporting tools:

- **Date Hierarchy**: Year -> Quarter -> Month -> Day
- **Complaint Hierarchy**: Service Category -> Complaint Group -> Complaint Type -> Descriptor
- **Location Hierarchy**: Borough -> City -> Incident Zip

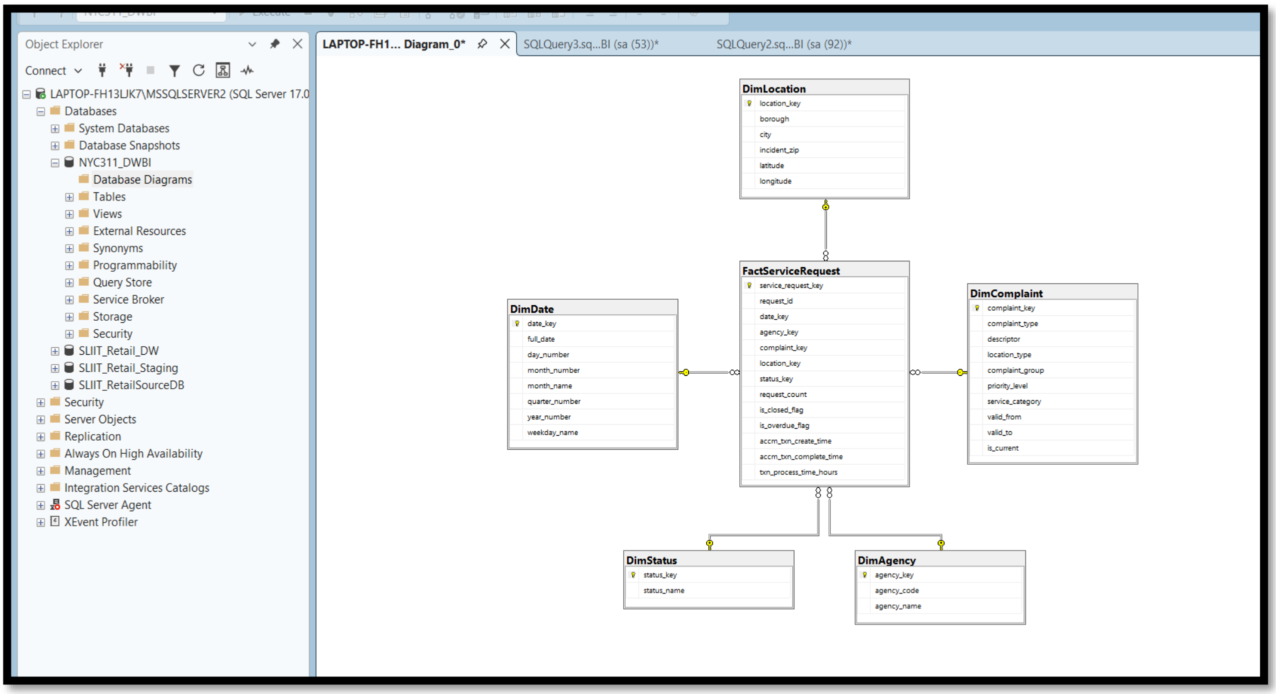
These hierarchies improve usability and enhance analytical capabilities.

4.6 Star Schema Diagram

The dimensional model is visually represented in the above star schema diagram. The FactServiceRequest table is located at the center and is connected to five-dimension tables: DimDate, DimAgency, DimComplaint, DimLocation, and DimStatus.

Each dimension provides descriptive attributes, while the fact table stores measurable data related to service requests. The relationships are maintained using foreign keys, ensuring referential integrity.

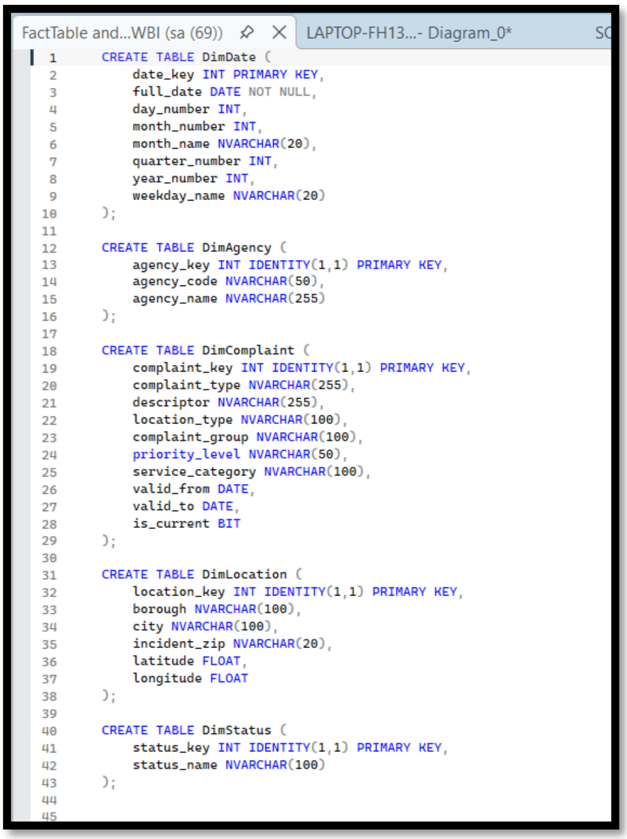
This structure supports efficient querying, aggregation, and reporting across multiple analytical perspectives.



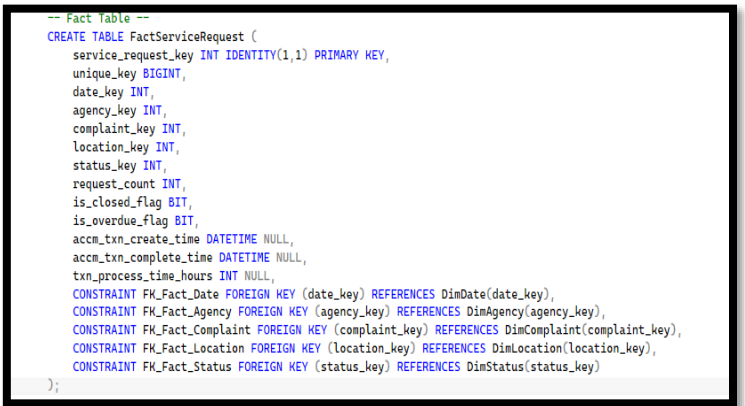
4.7 SQL Implementation

The data warehouse schema was implemented in SQL Server using structured tables with primary and foreign key constraints.

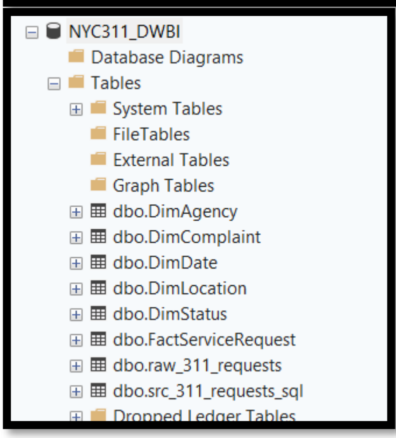
The implementation ensures data integrity and supports efficient joins between fact and dimension tables.




```sql
CREATE TABLE DimDate (
    date_key INT PRIMARY KEY,
    full_date DATE NOT NULL,
    day_number INT,
    month_number INT,
    month_name NVARCHAR(20),
    quarter_number INT,
    year_number INT,
    weekday_name NVARCHAR(20)
);

CREATE TABLE DimAgency (
    agency_key INT IDENTITY(1,1) PRIMARY KEY,
    agency_code NVARCHAR(50),
    agency_name NVARCHAR(255)
);

CREATE TABLE DimComplaint (
    complaint_key INT IDENTITY(1,1) PRIMARY KEY,
    complaint_type NVARCHAR(255),
    descriptor NVARCHAR(255),
    location_type NVARCHAR(100),
    complaint_group NVARCHAR(100),
    priority_level NVARCHAR(50),
    service_category NVARCHAR(100),
    valid_from DATE,
    valid_to DATE,
    is_current BIT
);

CREATE TABLE DimLocation (
    location_key INT IDENTITY(1,1) PRIMARY KEY,
    borough NVARCHAR(100),
    city NVARCHAR(100),
    incident_zip NVARCHAR(20),
    latitude FLOAT,
    longitude FLOAT
);

CREATE TABLE DimStatus (
    status_key INT IDENTITY(1,1) PRIMARY KEY,
    status_name NVARCHAR(100)
);
```



```sql
-- Fact Table --
CREATE TABLE FactServiceRequest (
    service_request_key INT IDENTITY(1,1) PRIMARY KEY,
    unique_key BIGINT,
    date_key INT,
    agency_key INT,
    complaint_key INT,
    location_key INT,
    status_key INT,
    request_count INT,
    is_closed_flag BIT,
    is_overdue_flag BIT,
    accm_txn_create_time DATETIME NULL,
    accm_txn_complete_time DATETIME NULL,
    txn_process_time_hours INT NULL,
    CONSTRAINT FK_Fact_Date FOREIGN KEY (date_key) REFERENCES DimDate(date_key),
    CONSTRAINT FK_Fact_Agency FOREIGN KEY (agency_key) REFERENCES DimAgency(agency_key),
    CONSTRAINT FK_Fact_Complaint FOREIGN KEY (complaint_key) REFERENCES DimComplaint(complaint_key),
    CONSTRAINT FK_Fact_Location FOREIGN KEY (location_key) REFERENCES DimLocation(location_key),
    CONSTRAINT FK_Fact_Status FOREIGN KEY (status_key) REFERENCES DimStatus(status_key)
);
```



4.8 Assumptions

The following assumptions were made during the design:

- Each row in the source dataset represents one service request
- The created date is used as the primary date reference
- Complaint classification attributes are derived from the Excel lookup file
- Missing completion timestamps indicate that the request is still open
- Processing time is calculated only for completed requests

Task 5: ETL Development

5.1 Overview

The Extract, Transform, Load (ETL) process was implemented using SQL Server Integration Services (SSIS) to populate the data warehouse from multiple data sources. The ETL process consists of two main stages: loading data into a staging area and transforming it into dimension and fact tables.

5.2 Data Sources Used

Two types of data sources were used:

- CSV File: NYC 311 Service Requests dataset
- Excel File: Complaint category lookup data

These sources were integrated to ensure data enrichment and support dimensional modeling.

5.3 ETL Architecture

The ETL process was divided into two SSIS packages:

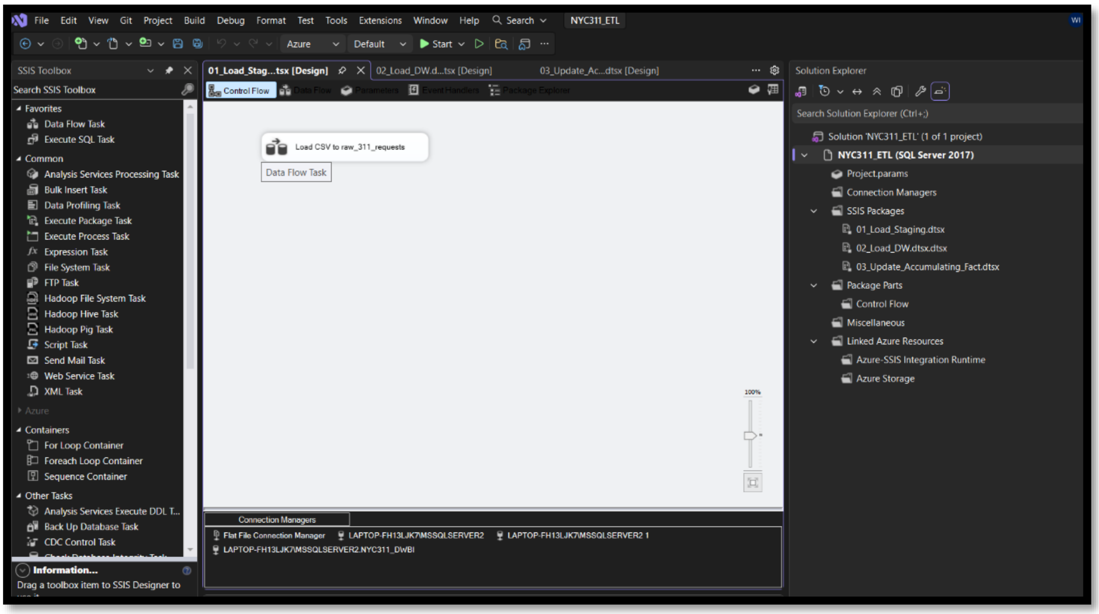
1. Staging Package (01_Load_Staging.dtsx)

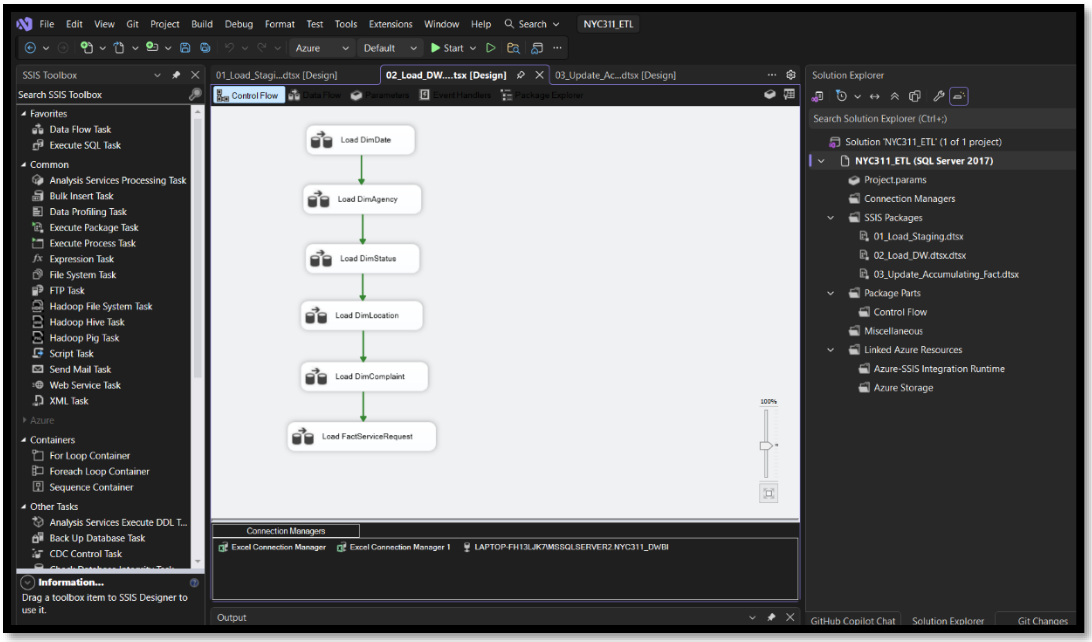
- Extracts raw data from the CSV file
- Loads data into the staging table raw_311_requests
- Uses Data Conversion to ensure proper data types

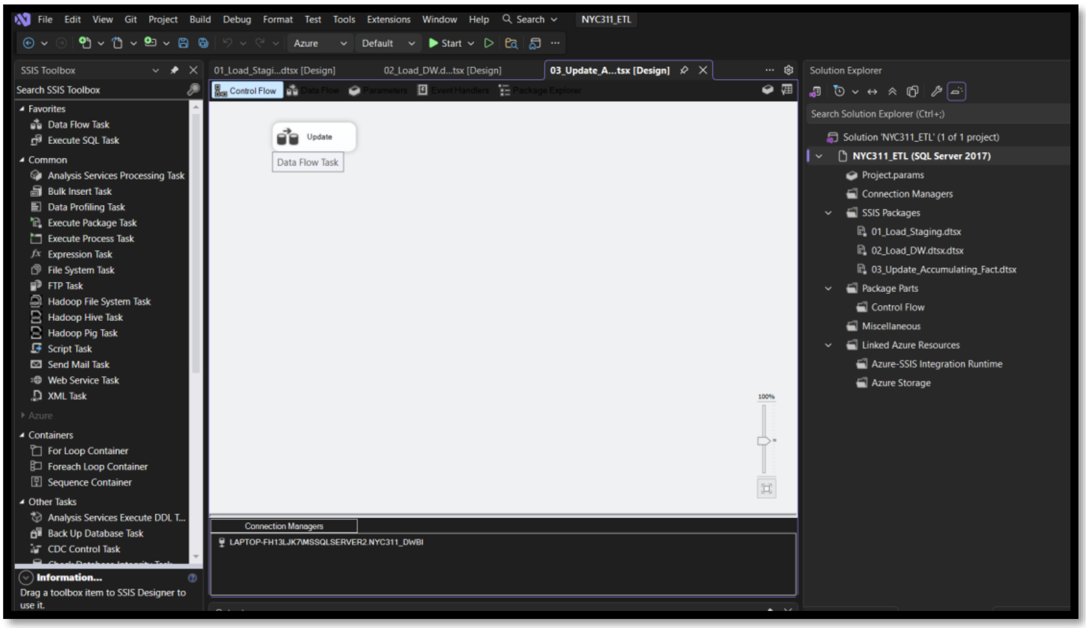
2. Data Warehouse Package (02_Load_DW.dtsx)

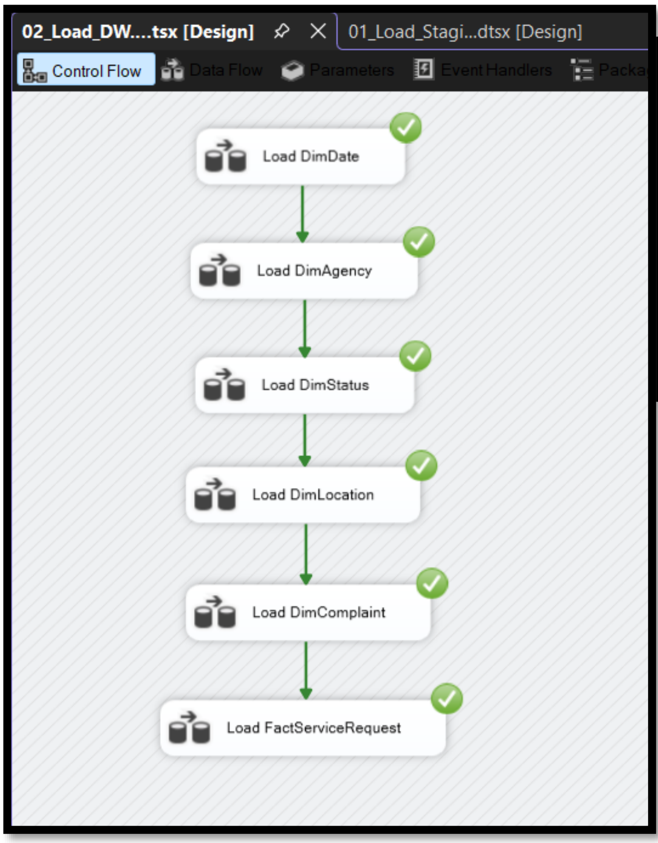
- Transforms and loads data into dimension and fact tables
- Uses Lookup transformations to generate surrogate keys

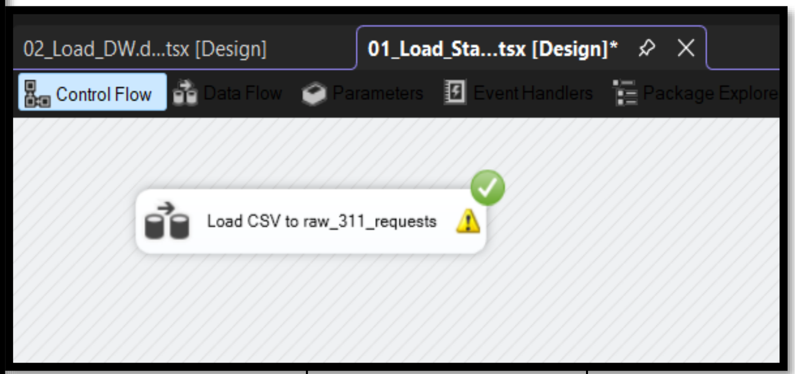
sources were integrated to ensure data enrichment and support dimensional modeling.


File Edit View Git Project Build Debug Format Test Tools Extensions Window Help Search NYC311_ETL
Azure Default Start
SSIS Toolbox
Search SSIS Toolbox
Favorites
Data Flow Task
Execute SQL Task
Common
Analysis Services Processing Task
Bulk Insert Task
Data Profiling Task
Execute Package Task
Execute Process Task
Expression Task
File System Task
FTP Task
Hadoop File System Task
Hadoop Hive Task
Hadoop Pig Task
Script Task
Send Mail Task
Web Service Task
XML Task
Azure
Containers
For Loop Container
Foreach Loop Container
Sequence Container
Other Tasks
Analysis Services Execute DDL T...
Back Up Database Task
CDC Control Task
Information...
Drag a toolbox item to SSIS Designer to
01_Load_Stag....tsx [Design]
02_Load_DW.d...tsx [Design]
03_Update_Ac...dtsx [Design]
Control Flow
Load CSV to raw_311_requests
Data Flow Task
100%
Connection Managers
Flat File Connection Manager
LAPTOP-FH13LJK7\MSSQLSERVER2
LAPTOP-FH13LJK7\MSSQLSERVER2 1
LAPTOP-FH13LJK7\MSSQLSERVER2.NYC311_DWBI
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'NYC311_ETL' (1 of 1 project)
NYC311_ETL (SQL Server 2017)
Project.params
Connection Managers
SSIS Packages
01_Load_Staging.dtsx
02_Load_DW.dtsx.dtsx
03_Update_Accumulating_Fact.dtsx
Package Parts
Control Flow
Miscellaneous
Linked Azure Resources
Azure-SSIS Integration Runtime
Azure Storage


File Edit View Git Project Build Debug Format Test Tools Extensions Window Help Search NYC311_ETL
Azure Default Start
SSIS Toolbox
Search SSIS Toolbox
01_Load_Stagi...dtsx [Design]
02_Load_DW....tsx [Design]
03_Update_Ac...dtsx [Design]
Control Flow
Load DimDate
Load DimAgency
Load DimStatus
Load DimLocation
Load DimComplaint
Load FactServiceRequest
100%
Connection Managers
Excel Connection Manager
Excel Connection Manager 1
LAPTOP-FH13LJK7\MSSQLSERVER2.NYC311_DWBI
Information...
Drag a toolbox item to SSIS Designer to use it.
Output
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'NYC311_ETL' (1 of 1 project)
NYC311_ETL (SQL Server 2017)
GitHub Copilot Chat
Solution Explorer
Git Changes


File Edit View Git Project Build Debug Format Test Tools Extensions Window Help Search NYC311_ETL
SSIS Toolbox
01_Load_Stagi...dtsx [Design]
02_Load_DW.d...tsx [Design]
03_Update_A...tsx [Design]
Control Flow
Update
Data Flow Task
100%
Connection Managers
LAPTOP-FH13LJK7\MSSQLSERVER2.NYC311_DWBI
Information...
Drag a toolbox item to SSIS Designer to use it.
Solution Explorer
Solution 'NYC311_ETL' (1 of 1 project)
NYC311_ETL (SQL Server 2017)





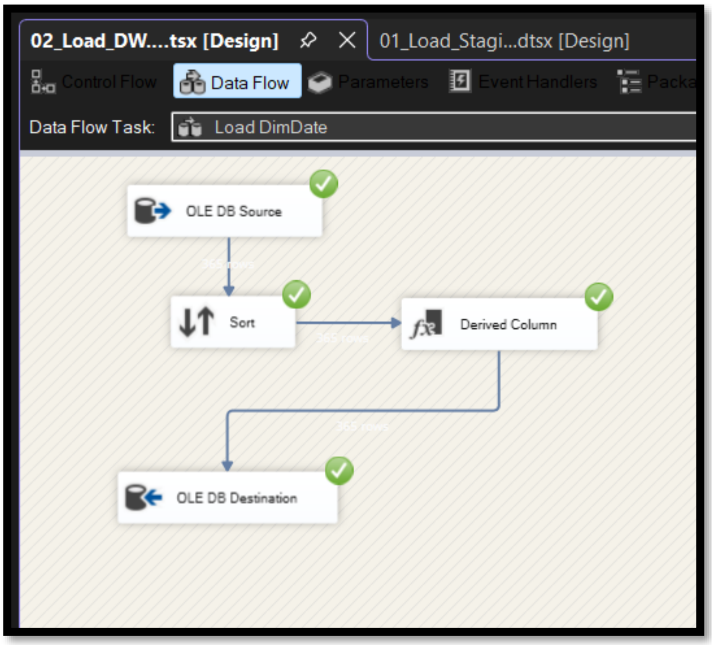
01_Load_Staging.dtsx

02_Load_DW.dtsx

5.4 Dimension Table Loading

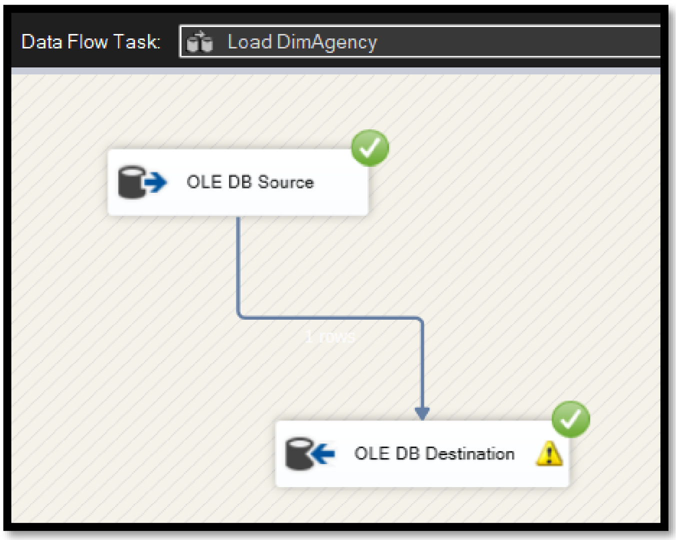
DimDate

- Extracted distinct dates from the dataset
- Generated a unique date_key
- Stored full date information



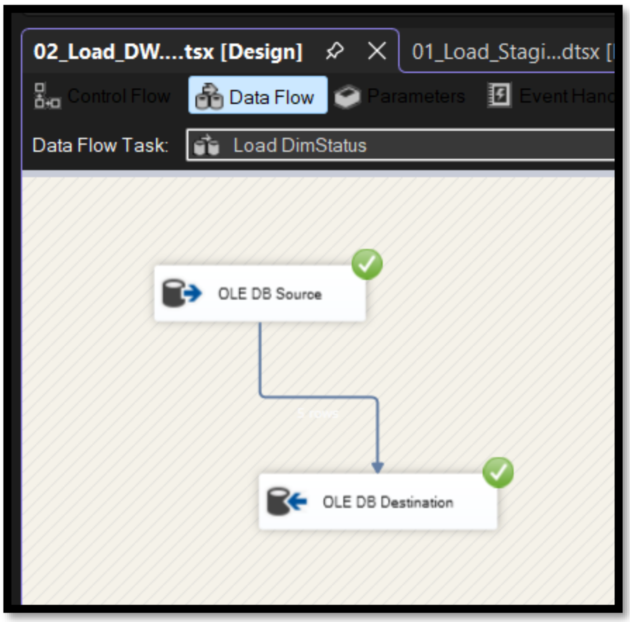
DimAgency

- Extracted unique agencies
- Applied grouping to avoid duplicates
- Stored agency code and name



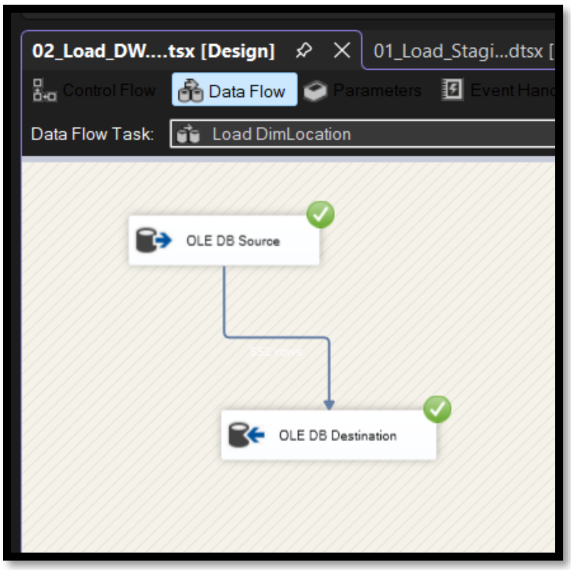
DimStatus

- Extracted distinct status values (e.g. Open, Closed)
- Stored as categorical dimension



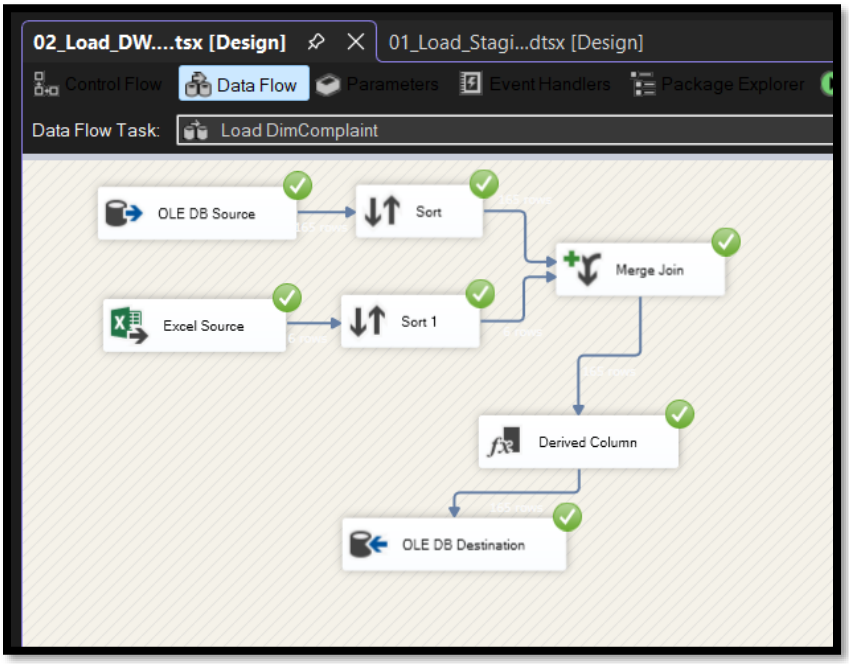
DimLocation

- Extracted location attributes such as borough, city, and zip code
- Handled missing or inconsistent geographic data



DimComplaint

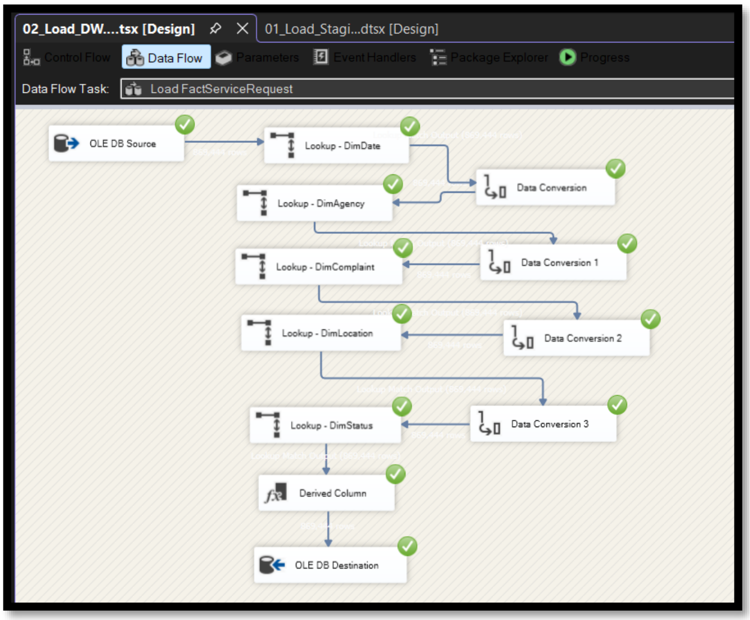
- Integrated Excel lookup data
- Used Merge Join to combine complaint information
- Generated enriched complaint categories



5.5 Fact Table Loading (FactServiceRequest)

The fact table was populated using:

- Multiple Lookup transformations:
  - DimDate
  - DimAgency
  - DimComplaint
  - DimLocation
  - DimStatus
- Each lookup replaced natural keys with surrogate keys
- Measures included:
  - request_count
  - is_closed_flag



5.6 Data Transformation Techniques Used

- Data Conversion: Ensured consistency between source and destination data types
- Lookup Transformation: Used for surrogate key generation
- Merge Join: Combined data from multiple sources
- Aggregation & Grouping: Removed duplicates in dimension tables

5.7 Loading Sequence

The ETL process follows a specific order to maintain referential integrity:

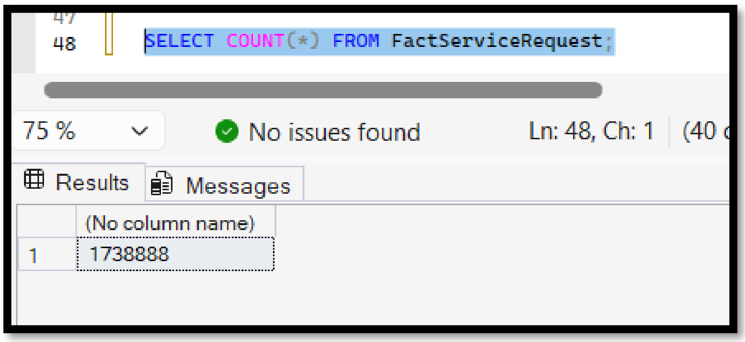
- DimDate
- DimAgency
- DimStatus
- DimLocation
- DimComplaint
- FactServiceRequest

5.8 Data Quality Handling

- Several data quality challenges were addressed:
- Null values were handled using filters and conditional logic
- Data type mismatches were resolved using Data Conversion
- Duplicate records were eliminated using grouping
- Lookup mismatches were handled by allowing non-matching rows

5.9 Execution Results

The ETL process was successfully executed, and data was loaded into all dimensions and fact tables.



Task 6: ETL development – Accumulating fact tables

To support the accumulating fact table requirement, the FactServiceRequest table was extended with three additional attributes:

- accm_txn_create_time
- accm_txn_complete_time
- txn_process_time_hours

The accm_txn_create_time attribute was initially populated during the fact table loading process using the current system timestamp. At that stage, accm_txn_complete_time was left as NULL because the service request was assumed to be incomplete.

A separate update dataset named completion_updates was then created to simulate the completion of transactions. This dataset contained the natural key of the fact table (request_id) together with a completion timestamp (completion_time).

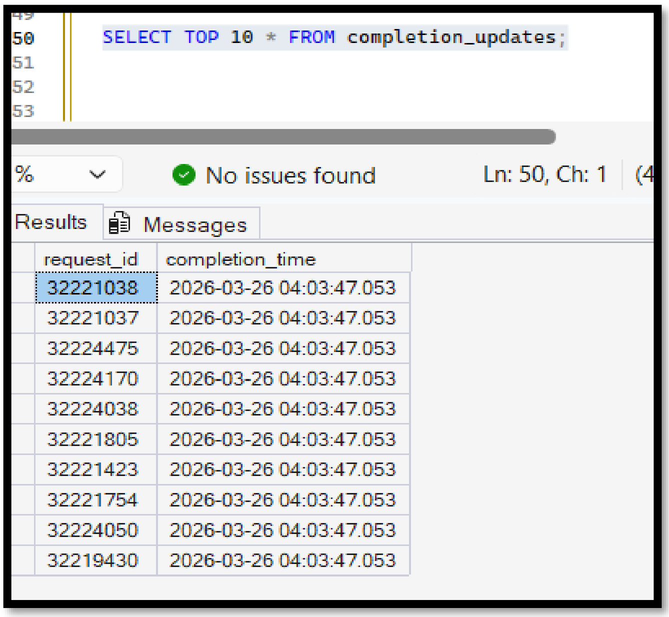
To process these updates, a separate SSIS package named 03_Update_Accumulating_Fact.dtsx was developed. The package extracted records from the completion_updates table and used an OLE DB Command transformation to update matching records in the FactServiceRequest table.

The update logic performed the following actions:

- updated accm_txn_complete_time
- calculated txn_process_time_hours as the difference in hours between accm_txn_complete_time and accm_txn_create_time

This process successfully demonstrated the implementation of an accumulating fact table, where a fact record is updated over time as the transaction progresses through its lifecycle.

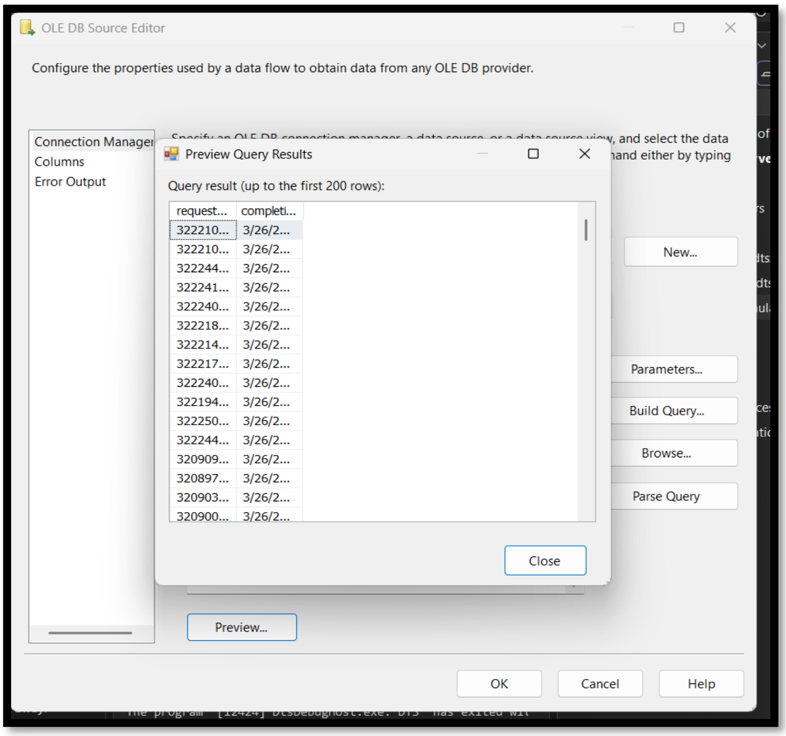
Completion update source table in SQL



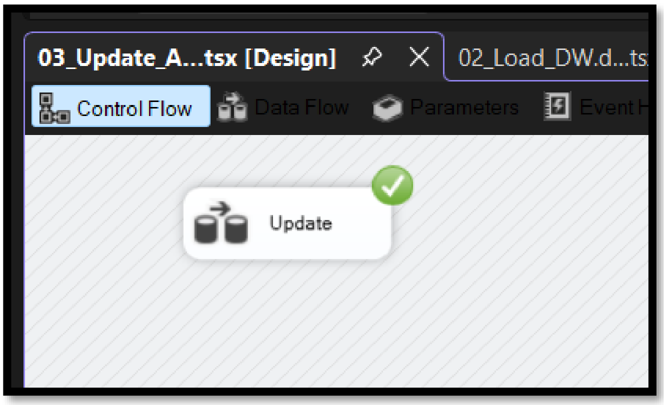
```sql
SELECT TOP 10 * FROM completion_updates;
```

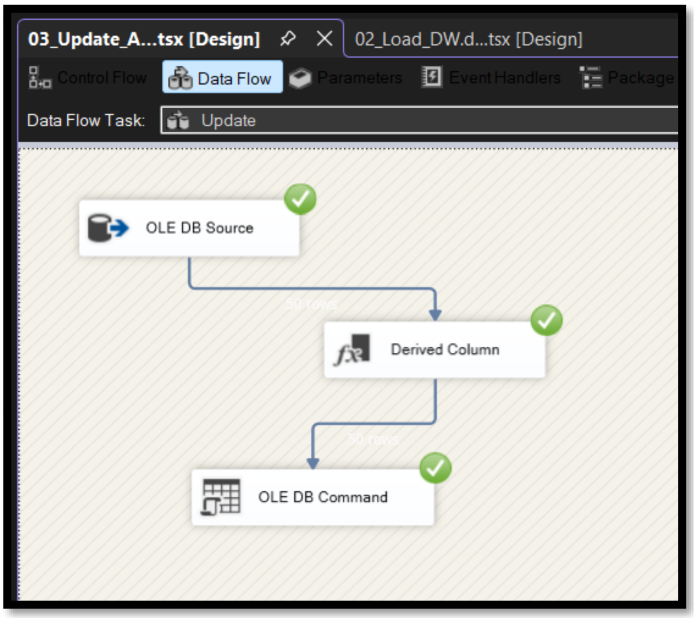
request_id	completion_time
32221038	2026-03-26 04:03:47.053
32221037	2026-03-26 04:03:47.053
32224475	2026-03-26 04:03:47.053
32224170	2026-03-26 04:03:47.053
32224038	2026-03-26 04:03:47.053
32221805	2026-03-26 04:03:47.053
32221423	2026-03-26 04:03:47.053
32221754	2026-03-26 04:03:47.053
32224050	2026-03-26 04:03:47.053
32219430	2026-03-26 04:03:47.053

OLE DB Source preview



SSIS Data Flow





SQL output after update

```sql
SELECT TOP 20
    request_id,
    accm_txn_create_time,
    accm_txn_complete_time,
    txn_process_time_hours
FROM FactServiceRequest
WHERE accm_txn_complete_time IS NOT NULL;
```

75 % No issues found

Results Messages

	request_id	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	32221038	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
2	32221037	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
3	32224475	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
4	32224170	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
5	32224038	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
6	32221805	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
7	32221423	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
8	32221754	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
9	32224050	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
10	32219430	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
11	32225009	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
12	32224465	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
13	32090970	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
14	32089701	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
15	32090378	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5
16	32090044	2026-03-25 23:03:47.053	2026-03-26 04:03:47.053	5

Number of updated accumulating fact records

```sql
SELECT COUNT(*) AS updated_rows
FROM FactServiceRequest
WHERE accm_txn_complete_time IS NOT NULL;
```

75 % No issues found

Results Messages

	updated_rows
1	1656